

Pawan Prakash

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Research Interests

Generative models (diffusion, flow matching, and stochastic interpolants) and reinforcement learning (GRPO) for crystals and molecules; equivariant and geometric deep learning; AI-accelerated materials and chemistry discovery; machine learning interatomic potentials; high-throughput screening and closed-loop workflows coupling ML, DFT, and experiment.

Education and Experience

Ph.D., Physics, University of Florida

2021–Present

Advisors: Richard G. Hennig, Peter J. Hirschfeld

Ph.D. thesis: Generative AI for the Accelerated Materials Discovery and Characterization

GPA: 4.00/4.00 (Expected Dec 2026)

Integrated B.S.–M.S., Physics, Indian Institute of Science Education and Research, Kolkata 2019

Master’s thesis: Solving the Hierarchy Problem Using Small Extra Dimensions

GPA: 7.98/10.00

Research Intern, Oak Ridge National Laboratory

May–Aug 2026

Mentors: Panchapakesan Ganesh, Addis Fuhr

Center for Nanophase Materials Sciences (CNMS)

Research Experience

Graduate Researcher, University of Florida

2022–Present

Advisors: Richard G. Hennig, Peter J. Hirschfeld

- **Guided diffusion for the discovery of new superconductors** (May 2024–May 2026)
 - Trained a foundation model on a large materials dataset on UF HPC (NVIDIA B200 GPUs) - a generative model for crystal structures.
 - Led a project spanning 3 universities and 8 PIs.
 - Spearheaded development of an adapter module to fine-tune the foundation model and used classifier-free guidance to do targeted generation with desired T_c .
 - Sampled 200,000 crystals, deduplicated to 34,027, yielding 773 DFT-validated candidates with $T_c > 5$ K (2.3% hit; $\sim 135\times$ over element-substitution baseline).
 - [Paper](#) published at npj Computational Materials. Selected Spotlight talk at NeurIPS AI4Mat, 2025.
- **Reinforcement Learning on the Discrete Composition Channel of a Crystal Generator: Validated Gains and Reward Hacking** (Jan 2026–Present)
 - First policy-gradient RL on the discrete species channel of a crystal generator: GRPO (PPO-clipped) over OMatG’s SDE+DFM rollouts with exact per-atom categorical log-probs; GPU-batched UMA energy-above-hull rewards on NVIDIA B200s.
 - Led the project end-to-end — originated the approach, implemented the training stack, and designed and ran all experiments.
 - More than tripled the stable-unique-novel yield over the pretrained prior (13.4% to 45.5%); $2\times$ the published state of the art under the independent LeMat-GenBench protocol.
 - Discovered and closed a reward-hacking mode that games the community S.U.N. metric; distilled a five-mode reward-hacking taxonomy for MLIP-based rewards.
 - [Preprint](#) available. Manuscript submitted to ICLR 2027.
- **Inverse design of 2D defect structures from a target electronic fingerprint** (ORNL CNMS, May–Aug 2026)
 - Built a DOS-conditioned diffusion generator, paired with an equivariant-GNN property checker, that designs point defects in 2D hosts to match a target electronic fingerprint; trained on ORNL’s DMX set (620 HSE06 calculations, 59 2D hosts).

- Conditioning on the target density of states cut round-trip error by 41% and recovered the exact defective composition in 70% of guided samples.
- Extended the generator to magnetic materials (MagDOSGen) by adding the magnetic condition to the graph: canonicalized DFT spin conventions and re-encoded the condition through a zero-init channel, turning an input that had degraded generation into one that improves it; the model now co-generates per-atom moments whose ordering beats chance across three independent runs.
- Manuscript under preparation. Code: github.com/Q-CAD/Spec2Struct/tree/edos-dmx2
- **Damage-aware restoration of STEM images** (ORNL CNMS, May–Aug 2026)
 - Diagnosed that beam-damage repair was erasing the very defects under study, and fixed it with a one-sided loss plus a self-setting confidence gate.
 - Cut invented atoms from 23 $\rightarrow$ 2 at light damage while preserving real defects at 0.9934 — as safe as not running the model.
 - Manuscript under preparation. Code: github.com/paparakash/stem_infilling
- **Benchmarking of Fast and Interpretable UF Machine Learning Potentials** (Jan 2022–Dec 2024)
 - Designed hyperparameter optimization and benchmarked UF3 against state-of-the-art MLPs (GAP, MTP, SNAP, NNP).
 - Demonstrated transferability by predicting melting points despite training only on solid-state data.
 - [Paper](#) accepted at Computational Materials Science; presented at the NeurIPS 2023 ML4PS workshop.
- **Open Materials Generation with Stochastic Interpolants (OMatG)** (Oct 2024–Jul 2025)
 - Co-developed and deployed an equivariant graph network for SI velocity-field learning in PyTorch Lightning; open-source.
 - Trained and tuned models, achieving SOTA on current benchmarks; implemented property-guided training for targeted generation.
 - [Paper](#) accepted at ICML 2025.
- **End-to-end AI-accelerated workflow for superconductor discovery** (Mar 2025–Jun 2025)
 - Designed and engineered the ML+DFT workflow to accelerate superconductor discovery.
 - Discovered 204 superconducting candidates from 1.22M materials, with one candidate experimentally realized and characterized.
 - [Paper](#) published at npj Computational Materials.
- **MolGuidance: a study of property guidance for molecule generation** (Mar 2024–Jul 2025)
 - Led debugging, hyperparameter studies, and clarified guidance-method selection.
 - First comparative study of guidance methods for conditional molecular generation.
 - [Paper](#) published at J. Chem. Inf. Model. and presented at the ICML 2025 GenBio Workshop.
- **Discovery of spin-crossover candidates with equivariant GNNs** (May 2023–Nov 2024)
 - Implemented metal-center attention and developed baselines.
 - Achieved 4 $\times$ improvement in identifying spin-crossover systems compared to baseline models.
 - [Paper](#) published at J. Chem. Theory Comput.

Publications and Manuscripts

Refereed published articles

- **P. Prakash**, J. B. Gibson, Zhongwei Li, Gabriele Di Gianluca, Juan Esquivel, Eric Fuemmeler, Benjamin Geisler, Jung Soo Kim, Adrian Roitberg, Ellad B. Tadmor, Mingjie Liu, Stefano Martiniani, Gregory R. Stewart, James J. Hamlin, Peter J. Hirschfeld, Richard G. Hennig. Guided Diffusion for the Discovery of New Superconductors. *npj Computational Materials* 12, 286 (2026). [Paper](#)
- **P. Prakash**, S. Dong, R. G. Hennig. Benchmarking of Fast and Interpretable UF Machine Learning Potentials. Accepted at *Computational Materials Science*. [Paper](#)
- J. Jin, C. Zeng, **P. Prakash**, E. B. Tadmor, A. Roitberg, R. G. Hennig, S. Martiniani, M. Liu. MolGuidance: Advanced Guidance Strategies for Conditional Molecular Generation with Flow Matching. *Journal of Chemical Information and Modeling* 66(15):8860–8874, 2026. [Paper](#)

- P. Höllmer, T. Egg, M. M. Martirosyan, E. Fuemmeler, Z. Shui, A. Gupta, **P. Prakash**, et al. Open Materials Generation with Stochastic Interpolants. *Proceedings of the 42nd International Conference on Machine Learning (ICML)*, 2025. [Paper](#)
- J. B. Gibson, A. C. Hire, **P. Prakash**, P. M. Dee, B. Geisler, J. S. Kim, Z. Li, J. J. Hamlin, G. R. Stewart, P. J. Hirschfeld, R. G. Hennig. Developing a Complete AI-Accelerated Workflow for Superconductor Discovery. *npj Computational Materials* 12, 95 (2026). [Paper](#)
- A. Albavera-Mata, **P. Prakash**, J. B. Gibson, E. Fonseca, S. Ren, X.-G. Zhang, S. B. Trickey, R. G. Hennig. Discovery of Spin-Crossover Materials with Equivariant Graph Neural Networks and Relevance-Based Classification. *Journal of Chemical Theory and Computation* 21(8):3913–3921, 2025. [Paper](#)

Manuscripts under review

- **P. Prakash**, P. Höllmer, A. Fuhr, P. Hirschfeld, G. Panchapakesan, S. Martiniani, R. Hennig. Reinforcement Learning on the Discrete Composition Channel of a Crystal Generator: Validated Gains and Reward Hacking (OMatGRPO). Submitted to ICLR 2027; [preprint](#) on arXiv (Oct 2026); code on [github](#).

Talks and Posters

- Generative AI for Accelerated Materials Discovery and Characterization — CNMS, ORNL, Oak Ridge, 2026. (Invited talk)
- OMatGRPO: RL fine-tuning of a stochastic-interpolant crystal generator — MRS spring meet, Hawaii, 2026.
- Guided diffusion for the discovery of new superconductors — AI4Mat - NeurIPS, San Diego, 2025. (Spotlight Talk)
- Guided diffusion for the discovery of new superconductors — AI4SC workshop, University of Florida, 2025. (Invited Talk)
- Open Materials Generation with Stochastic Interpolants (OMatG) — ICML, Vancouver, BC, 2025. (Poster)
- MolGuidance: a study of property guidance for molecule generation — GenBio workshop ICML, Vancouver, BC, 2025. (Poster)
- Guided diffusion for the discovery of new superconductors — AI4Chemistry Summit, NYU, 2025. (Poster)
- Guided diffusion for the discovery of new superconductors — IAIFI summer school, Harvard, 2025. (Talk)
- Guided diffusion for the discovery of new superconductors — APS March Meeting, Anaheim, CA, 2025. (Talk)
- Guided diffusion for the discovery of new superconductors — Sanibel symposium, St. Augustine Beach, FL, 2025. (Poster)
- Benchmarking of Fast and Interpretable UF Machine Learning Potentials — MRS Spring Meeting, Seattle, 2024. (Poster)
- Benchmarking of Fast and Interpretable UF Machine Learning Potentials — ML4PS - NeurIPS, New Orleans, 2023. (Poster)

Awards and Honors

- Dr. Emmanuel and Dora G. Partheniades Physics Award, 2025.
- IHEPAA fellowship, University of Florida, 2021.
- INSPIRE Scholarship, 2014-19.

Open-Source and Datasets

- Guided diffusion (open-source): Guided diffusion for the discovery of new superconductors. <https://github.com/paparakash/GuidedMatDiffusion>
- OMatGRPO (open-source): RL fine-tuning of a crystal generator. <https://github.com/paparakash/OMatGRPO>
- OMatG (open-source): stochastic-interpolant materials generation in PyTorch Lightning. <https://github.com/FERMat-ML/OMatG>

Technical Skills

AI/ML: PyTorch, PyTorch Lightning, CUDA Graphs, AMP/mixed precision.

Scientific computing: Python, MPI, Slurm, High-performance computing (HPC).

Materials/physics: VASP, Quantum ESPRESSO, electron-phonon workflows, ASE, MLIPs (e.g., MatterSim).

Tooling: NumPy, SciPy, Pandas, Dask, Weights & Biases.

HPC: NVIDIA A100/B200, Distributed training (DDP/FSDP), profiling, memory optimization.

Teaching and Mentoring

- Teaching Assistant, Physics I lab, University of Florida, Spring, 2021 — lab lectures, grading.
- Teaching Assistant, Physics II discussion classes, University of Florida, fall, 2021 — lectures, quiz preparation, grading.